

PRODUCTION FUNCTIONS PROBLEM SET—SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

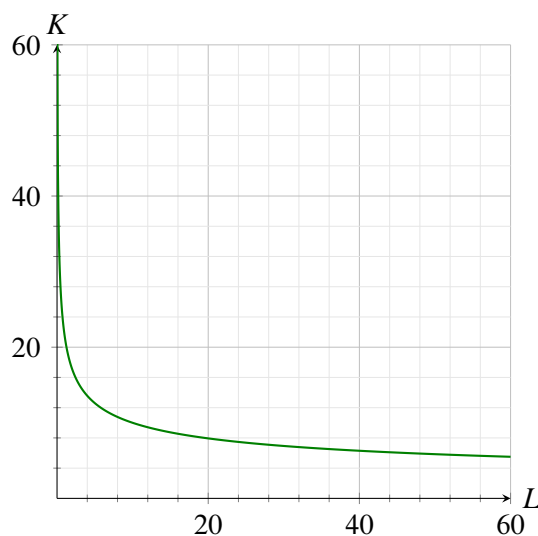
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PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Problem 1

(a) $F(L, K) = LK^3$



Marginal products:

$$MP_L = K^3, \quad MP_K = 3LK^2.$$

The marginal product of Labor is constant with respect to L , while the marginal product of capital is constant with respect to K .

For the $MRTS$ of K for L , we get

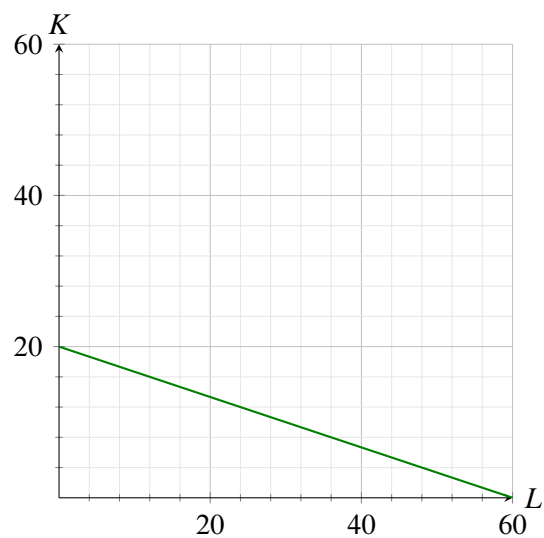
$$MRTS = \frac{K^3}{3LK^2} = \frac{K}{3L}.$$

Checking returns to scale, we multiply all inputs by a factor $t > 1$:

$$F(tL, tK) = (tL)(tK)^3 = t^4 LK^3 = t^4 F(L, K),$$

thus this production function exhibits increasing returns to scale.

(b) $F(L, K) = L + 3K$



Marginal products:

$$MP_L = 1, \quad MP_K = 3.$$

Both marginal products are constant.

For the $MRTS$ of K for L , we get

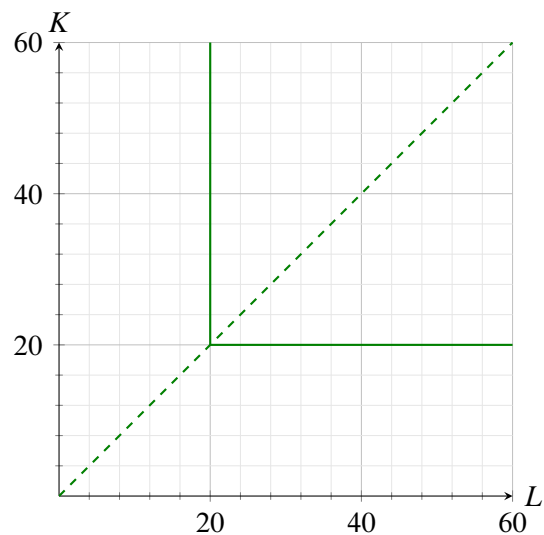
$$MRTS = \frac{1}{3} = .$$

Checking returns to scale, we multiply all inputs by a factor $t > 1$:

$$F(tL, tK) = tL + 3tK = tF(L, K),$$

thus this production function exhibits constant returns to scale.

(c) $F(L, K) = (\min\{L, K\})^{\frac{1}{3}}$



Marginal products:

$$MP_L = \begin{cases} 0 & \text{if } L \geq K \\ \frac{1}{3}L^{-\frac{2}{3}} & \text{if } L < K \end{cases}$$

$$MP_K = \begin{cases} 0 & \text{if } L \leq K \\ \frac{1}{3}K^{-\frac{2}{3}} & \text{if } L > K \end{cases}$$

For the *MRTS*:

$$MRTS = \begin{cases} 0 & \text{if } L > K \\ \infty & \text{if } L < K \end{cases},$$

and it is undefined at the kink points where $L = K$.

Checking returns to scale, we multiply all inputs by a factor $t > 1$:

$$F(tL, tK) = (\min\{tL, tK\})^{1/3} = (t \min\{L, K\})^{1/3} = t^{1/3} \min\{L, K\} < tF(L, K),$$

thus this production function exhibits decreasing returns to scale.

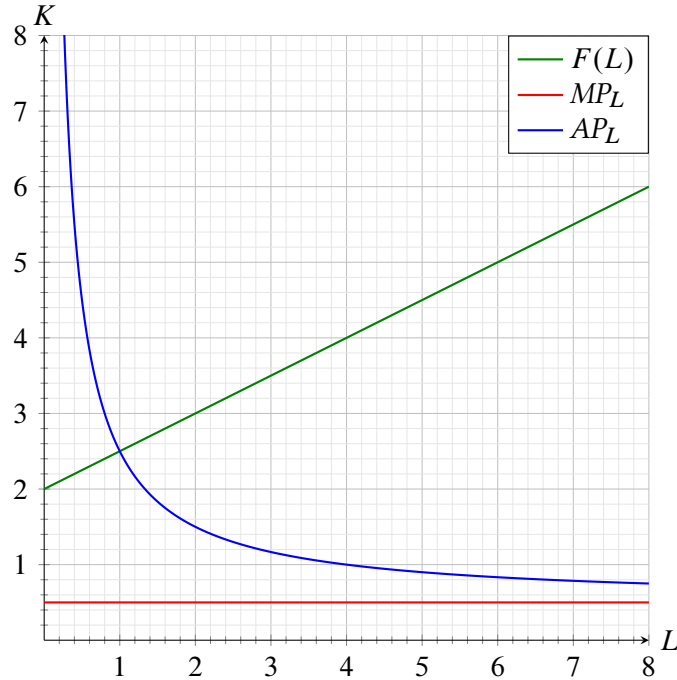
Problem 2

- (a) We multiply all inputs by a factor $t > 1$:

$$F(tL, tK) = tL/2 + \sqrt{tK} < t(L/2 + \sqrt{K}) = tF(K, L),$$

thus, it exhibits decreasing returns to scale.

- (b) $MP_L = 1/2$, thus, it is constant.
- (c) The short run production function when $K = 4$ is given by $F(L) = L/2 + \sqrt{4} = L/2 + 2$. See the graph below.
- (d) The short run marginal product of labor is $MP_L = 1/2$ and the $AP_L = 1/2 + 2/L$. See the graph below.



Problem 3

The firm Muppets Inc., has two plants for producing widgets, one in Sesame Street and another one in Fraggle Rock. The Sesame Street plant produces according to

$$F_S(x_1, x_2) = \min\{x_1, 2x_2\},$$

and the Fraggle Rock plant produces according to

$$F_F(x_1, x_2) = \min\{2x_1, x_2\},$$

where x_1 and x_2 are inputs of production.

- See the diagram below. The isoquants for 40 output units for the Sesame Street and Fraggle Rock plants are red and blue, respectively.
- At the Sesame street plant, we need the input combination (20, 10) and at the Fraggle Rock plant we need the input combination (10, 20) to produce 20 widgets at each. Thus, bundle A is given by $A = (30, 30)$
- At the Sesame Street plant, now we need the input combination (10, 5), and in the Fraggle Rock plant we need the input combination (15, 30). Input combination B is then $B = (25, 35)$.
- At the Sesame Street plant, we need the input combination (30, 15), and in the Fraggle Rock plant we need the input combination (5, 10). Input combination C is then $C = (35, 25)$.
- See the graph below. The isoquant is for the combined plants is depicted in dark green with a dashed line.

